LITTLE HUB

A Mini Project Report

Submitted for partial fulfillment of the requirement for the award of the degree of

MASTER OF SCIENCE IN INFORMATION TECHNOLOGY

Submitted by

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CERTIFFICATE

This is to certify that the Mini-project entitled

LITTLE HUB

Being Submitted to

ST.JOSEPH'S COLLEGE OF ARTS AND SCIENCE (AUTONOMOUS)
CUDDALORE-1
(Affiliated to Annamalai University Chidambaram)

Submitted by

SAIDHA FATHIMA K F (A24MIT21)

For the Partial Fulfillment for the award of degree of

MASTER OF SCIENCE IN INFORMATION TECHNOLOGY

Is a Bonafide record of work carried out by her, Under my guidance and supervision.

INTERNAL GUIDE **HEAD OF THE DEPARTMENT**

Submitted for the viva-voce examination on _____

Examiners:

1. _____

2. _____

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SAIDHA FATHIMA K F

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ABSTRACT

The LittleHUB project is a comprehensive web-based platform designed to facilitate the management, distribution, and consumption of digital educational resources, including books and notes. It provides a secure, interactive, and user-friendly environment for students, allowing them to easily browse, purchase, and access educational content from anywhere. The system implements role-based access control, differentiating between regular users and administrators to maintain security and organization.

Regular users can register, log in, view available books and notes, make purchases, and download content in multiple formats, including PDF. The platform tracks user activity to ensure transparency and personalized content management. Administrators have a dedicated dashboard that provides real-time insights into total users, books, notes, and purchase records. They can perform full CRUD operations on users, books, and notes, upload cover images and PDFs, and monitor platform activity.

The project leverages PHP for server-side logic, MySQL for database management, and HTML, CSS, and JavaScript for a responsive and interactive front-end design. File upload management, secure login authentication, and error handling are implemented to ensure smooth operation and data integrity. LittleHUB enhances the accessibility and organization of educational content, making it an efficient tool for digital learning while streamlining administrative tasks.

PROJECT DESCRIPTION

LittleHub is a comprehensive web-based platform designed to manage, organize, and provide access to e-books and academic notes efficiently. The project focuses on creating a digital library system that is easy to navigate for both end-users (students, learners) and administrators. It eliminates the need for physical storage of books and notes, providing instant digital access to educational resources.

Purpose of the Project:

The primary goal of LittleHub is to provide a centralized system for digital learning resources. Users can browse, purchase, and download books and notes securely, while administrators can manage users, content, and transactions efficiently. This system reduces manual administrative efforts, ensures the integrity of data, and provides a secure and user-friendly environment for educational content management.

Scope of the Project:

- For students and users:
- Access a variety of e-books and notes.
- Purchase or download resources securely.
- Maintain a personal dashboard to track purchased items and download history.

For administrators:

- Manage users, including adding, editing, and deleting accounts.
- Upload, update, or remove books and notes.
- Monitor purchases and user activity logs for reporting purposes.
- The system ensures role-based access, where admin functionalities are restricted to authorized personnel.

Key Functionalities:

User Module:

- Registration, login, and secure session management.

- Personalized dashboard displaying purchased resources.
- Ability to view/download purchased books and notes.
- Activity logging for every download or view.

Admin Module:

- CRUD operations for users, books, and notes.
- Upload and manage book cover images, PDFs, and notes files.
- Monitor purchases, track downloads, and generate activity reports.

Database Management:

- Uses MySQL relational database to store structured data for users, books, notes, and transactions.
- Maintains foreign key relationships to ensure consistency.

Security & Authentication:

- Role-based access ensures only admins can manage resources.
- Passwords stored securely using hashing.
- Sessions prevent unauthorized access.

User Interface:

- Clean, responsive, and intuitive design using HTML5, CSS3, and JavaScript.
- Font Awesome icons improve usability and visual appeal.

File Management:

- Secure upload and storage of book cover images, PDFs, and notes files.
- Handles file updates safely to prevent data loss or overwrite errors.

Benefits:

- Efficient access to learning resources anytime and anywhere.
- Reduces dependency on physical books and notes.
- Helps administrators manage educational resources with minimal effort.

HARDWARE AND SOFTWARE REQUIREMENTS

HARDWARE REQUIREMENTS

SYSTEM / LAPTOP : Intel Core i5 (12th Gen) 64-bit

STORAGE : 512 GB SSD

DISPLAY : 15.6” LED Laptop Screen

INPUT DEVICES : Built-in Keyboard & Touchpad (External Mouse Optional)

RAM : 8 GB DDR4

INTERNET CONNECTION : Broadband / Wi-Fi

SOFTWARE REQUIREMENTS

OPERATING SYSTEM : Windows 11 Professional / Windows 10 (64-bit)

PROCESSOR : Intel Core i5 (12th Gen) 64-bit

FRONT END : HTML5, CSS3, JavaScript

BACK END : PHP 7.4+

DATABASE : MySQL

WEB SERVER : (WAMP / XAMPP)

IDE / EDITOR : NetBeans / Visual Studio Code

BROWSER : Google Chrome / Microsoft Edge / Mozilla Firefox

FILE STORAGE : Local File System for Book Covers & Notes

ADDITIONAL TOOLS : phpMyAdmin for Database Management

SOFTWARE DESCRIPTION

Project Title: LittleHub

Type of Application: Web-Based Book and Notes Management System

Platform Used: WAMP (Windows, Apache, MySQL, PHP)

Front-End Technologies: HTML5, CSS3, JavaScript, Font Awesome

Back-End Technologies: PHP (Server-side scripting), MySQL (Database)

Server: Apache (via WAMP)

IDE Used: Visual Studio Code / Sublime Text / Notepad++

Operating System: Windows 10 / 11

1. User Management Module

Purpose:

Manages all users in the system and controls access levels.

Features:

User Registration: New users can create accounts with their name, email, and password.

User Login/Authentication: Validates credentials and establishes sessions.

Role-Based Access: Differentiates between admin and regular users.

Profile Management: Users can update their personal information.

Activity Logging: Tracks user actions like book downloads or notes viewing.

Admin Functionality:

Add, edit, and delete users.

View registered user list with detailed info.

Manage user roles and monitor activity.

2. Book Management Module

Purpose:

Handles all book-related functionalities including uploading, editing, and purchasing.

Features:

CRUD Operations: Admin can Create, Read, Update, and Delete book records.

File Uploads: Securely upload cover images and PDF files.

Metadata Storage: Stores title, author, price, cover image path, and PDF path.

Purchase Control: Only purchased books can be downloaded by users.

Search & Filter: Users can browse books by title or author.

Admin Functionality:

Manage all books in the system.

Update book details including files.

Remove outdated or irrelevant books.

3. Notes Management Module

Purpose:

Manages academic notes and additional learning resources.

Features:

CRUD Operations: Admin can add, edit, delete, or view notes.

File Uploads: Upload notes in PDF format securely.

Metadata Storage: Stores title, subject, description, uploaded by, and upload timestamp.

User Access: Users can view or download notes as permitted.

Admin Functionality:

Monitor all uploaded notes.

Edit or remove notes if necessary.

Maintain organized notes repository for users.

4. Purchases Module

Purpose:

Tracks all book purchases and controls access to downloadable content.

Features:

Purchase Recording: Logs each book purchase with user ID, book ID, and timestamp.

Download Restriction: Ensures only purchased books can be downloaded.

History Tracking: Users can view their purchase history.

Admin Functionality:

View all purchases in a tabular format.

Track user purchases and generate reports.

5. Admin Dashboard

Purpose:

Centralized control panel for administrators to manage the system efficiently.

Features:

Summary Statistics: Displays total users, books, notes, and purchases.

Quick Navigation: Provides links to Users, Books, Notes, and Purchases management.

Actionable Tables: Allows admins to perform CRUD operations directly from tables.

Activity Monitoring: Track system usage and user activities.

6. Security & Access Control Module

Purpose:

Ensures the system is secure and only authorized users can perform sensitive actions.

Features:

Session Management: Keeps users logged in securely; prevents unauthorized access.

Password Hashing: Encrypts passwords before storing in the database.

Role-Based Access Control: Restricts access to admin-only features.

File Upload Security: Ensures safe handling of PDFs and images to prevent overwriting or malicious files.

7. User Interface Module

Purpose:

Provides an intuitive and responsive interface for users and admins.

Features:

Responsive Design: Works on desktops, tablets, and mobile devices.

Consistent Theme: Uses HTML5, CSS3, and Font Awesome icons for a uniform look.

Interactive Forms and Tables: Simplifies CRUD operations for admins.

Clear Navigation: Navbar links for easy movement across modules.

8. Database Module

Purpose:

Stores all structured information securely and efficiently.

Features:

MySQL Database: Relational tables for Users, Books, Notes, and Purchases.

Foreign Key Constraints: Ensures relational integrity.

Query Support: Supports CRUD operations and statistical queries.

File Path Storage: Tracks uploaded files for books and notes.

9. File Management Module

Purpose:

Handles storage and retrieval of uploaded files.

Features:

Organized Folders: books_pdf/, cover/, notes_pdf/ for storing files.

Safe Updates: Allows replacing files without data loss.

Path Tracking: Stores file paths in the database for quick access.

DATABASE STRUCTURE

Database Name: lithub_db

1. Users Table

Stores all registered users including admin and students.

Column	*Type*	*Attributes*	*Description*
<u>user_id</u>	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each user
name	VARCHAR(100)	NOT NULL	User full name
email	VARCHAR(100)	UNIQUE, NOT NULL	User email
password	VARCHAR(255)	NOT NULL	Hashed password (MD5/other)
role	ENUM('admin','user')	NOT NULL	User role
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation date

2. Books Table

Stores all book details.

Column	*Type*	*Attributes*	*Description*
<u>book_id</u>	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each book
title	VARCHAR(255)	NOT NULL	Book title
author	VARCHAR(150)	NOT NULL	Book author
price	DECIMAL(10,2)	NOT NULL	Book price
cover	VARCHAR(255)	NULL	Cover image filename
pdf	VARCHAR(255)	NULL	PDF file filename
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date book added

3. Notes Table

Stores notes uploaded by users/admin.

Column	*Type*	*Attributes*	*Description*
<u>note_id</u>	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each note
subject	VARCHAR(150)	NOT NULL	Subject of the note
title	VARCHAR(255)	NOT NULL	Note title

description	TEXT	NULL	Optional description
file_path	VARCHAR(255)	NOT NULL	Uploaded file path
uploaded_by	INT	NOT NULL	User who uploaded the note
uploaded_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Upload date

Foreign Key: uploaded_by → users.user_id

4. Purchases Table

Tracks which user purchased which book.

Column	***Type***	***Attributes***	***Description***
purchase_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique purchase ID
user_id	INT	NOT NULL	User who purchased
book_id	INT	NOT NULL	Book purchased
purchased_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Purchase date

Foreign Keys:

user_id → users.user_id

book_id → books.book_id

Relationships:

Users → Purchases: One-to-Many (one user can have many purchases)

Books → Purchases: One-to-Many (one book can be purchased multiple times)

Users → Notes: One-to-Many (one user can upload multiple notes)

SOURCE CODE

Db.php

```
<?php
$servername = "localhost";
$username = "root";
$password = ""; // default WAMP password
$dbname = "lithub_db";

$conn = mysqli_connect($servername, $username, $password, $dbname);

if (!$conn) {
   die("Connection failed: " . mysqli_connect_error());
}
?>
```

Register.php

```
<?php
session_start();
include 'db.php';
$message = "";
if ($_SERVER['REQUEST_METHOD'] == 'POST') {
   $name = mysqli_real_escape_string($conn, $_POST['name']);
   $email = mysqli_real_escape_string($conn, $_POST['email']);
   $password = md5($_POST['password']); // hash using md5
   $role = 'user';

   // Check if email exists
   $check = mysqli_query($conn, "SELECT * FROM users WHERE email='$email'");
   if(mysqli_num_rows($check) > 0){
      $message = "Email already registered!";
   } else {
      $insert = mysqli_query($conn, "INSERT INTO users (name,email,password,role) VALUES
('$name','$email','$password','$role')");
      if($insert){
         $message = "Registration successful! <a href='login.php'>Login Now</a>";
      } else {
         $message = "Registration failed. Please try again.";
      }
   }
}
?>
<!DOCTYPE html>
<html lang="en">
<body>
   <div class="header-logo">
      <i class="fas fa-book-open"></i>
```

```
        <span>LittleHUB</span>
    </div>
<div class="form-container">
    <h2>Register</h2>
    <?php if($message != "") { echo "<p class='message'>$message</p>"; } ?>
    <form action="" method="POST">
        <input type="text" name="name" placeholder="Full Name" required>
        <input type="email" name="email" placeholder="Email" required>
        <input type="password" name="password" placeholder="Password" required>
        <button type="submit" class="btn btn-primary">Register</button>
    </form>
    <p>Already have an account? <a href="login.php">Login Here</a></p>
</div>
</body>
</html>
```

Login.php

```
<?php
session_start();
include 'db.php';
$message = "";
if ($_SERVER['REQUEST_METHOD'] == 'POST') {
    $email = mysqli_real_escape_string($conn, $_POST['email']);
    $password = $_POST['password'];
    $result = mysqli_query($conn, "SELECT * FROM users WHERE email='$email'");
    if(mysqli_num_rows($result) > 0){
        $user = mysqli_fetch_assoc($result);
        if($user['password'] === md5($password)){
            // Set session variables
            $_SESSION['user_id'] = $user['user_id'];
            $_SESSION['user_name'] = $user['name'];
            $_SESSION['user_role'] = $user['role'];
            $_SESSION['user_logged_in'] = true;

            // Redirect based on role
            if($user['role'] === 'admin'){
                header("Location: admin/admin_dashboard.php");
            } else {
                header("Location: dashboard.php");
            }
            exit();
        } else {
            $message = "Incorrect password!";
        }
    } else {
        $message = "No account found with this email!";
```

```
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Login - LittleHUB</title>
<link rel="stylesheet" href="css/login_register.css">
<link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/5.15.4/css/all.min.css">
</head>
<body>
<div class="header-logo">
    <i class="fas fa-book-open"></i>
    <span>LittleHUB</span>
</div>
<div class="form-container">
    <h2>Login</h2>
    <?php if($message != "") { echo "<p class='message'>$message</p>"; } ?>
    <form action="" method="POST">
        <input type="email" name="email" placeholder="Email" required>
        <input type="password" name="password" placeholder="Password" required>
        <button type="submit" class="btn btn-primary">Login</button>
    </form>
    <p>Don't have an account? <a href="register.php">Register Here</a></p>
</div>
</body>
</html>
```

Index.php

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>LittleHUB - Your Digital Learning Hub</title>
    <link rel="stylesheet" href="css/style.css">
    <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/5.15.4/css/all.min.css">
</head>
<body>
<nav class="navbar" id="navbar">
    <div class="nav-container">
        <div class="nav-brand">
```

```
            <i class="fas fa-book-open"></i>
            <span>LittleHUB</span>
        </div>
        <div class="nav-actions">
            <button class="btn btn-outline"
onclick="window.location.href='login.php'">Login</button>
            <button class="btn btn-primary"
onclick="window.location.href='register.php'">Register</button>
        </div>
    </div>
</nav>
<!-- Main Content -->
<main id="mainContent">
    <section id="home" class="section active">
        <div class="hero">
            <div class="hero-content">
                <h1 class="hero-title">Welcome to LittleHUB</h1>
                <p class="hero-subtitle">Your comprehensive digital learning hub with books, notes,
and more</p>
                <div class="hero-actions">
                    <button class="btn btn-primary btn-lg"
onclick="window.location.href='#books'">Explore Books</button>
                    <button class="btn btn-outline btn-lg"
onclick="window.location.href='#notes'">Download Notes</button>
                </div>
            </div>
            <div class="hero-image">
                <img src="images/library.jpeg" alt="Library">
            </div>
        </div>
        <div class="features">
            <div class="container">
                <h2 class="section-title">Our Features</h2>
                <div class="features-grid">
                    <div class="feature-card" onclick="window.location.href='#books'">
                        <div class="feature-icon"><i class="fas fa-book-reader"></i></div>
                        <h3>Online Reading</h3>
                        <p>Access thousands of books online with our advanced reading interface</p>
                    </div>
                    <div class="feature-card" onclick="window.location.href='#store'">
                        <div class="feature-icon"><i class="fas fa-shopping-cart"></i></div>
                        <h3>Buy Books</h3>
                        <p>Purchase physical and digital books from our extensive catalog</p>
                    </div>
                    <div class="feature-card" onclick="window.location.href='#notes'">
                        <div class="feature-icon"><i class="fas fa-download"></i></div>
```

```
                <h3>College Notes</h3>
                <p>Download comprehensive notes for various college subjects</p>
              </div>
            </div>
          </div>
        </div>
    </section>
</main>
</body>
</html>
```

style.php

```
/* ===== RESET & BASE ===== */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}
body {
    font-family: 'Inter', Arial, sans-serif;
    background: #f7f7f9;
    color: #333;
    line-height: 1.6;
    min-height: 100vh;
    scroll-behavior: smooth;
}
/* ===== NAVBAR ===== */
.navbar {
    position: fixed;
    top: 0;
    left: 0;
    width: 100%;
    background: rgba(255, 255, 255, 0.95);
    backdrop-filter: blur(10px);
    border-bottom: 1px solid rgba(0, 0, 0, 0.08);
    z-index: 1000;
}
.nav-container {
    max-width: 1200px;
    margin: auto;
    padding: 0 20px;
    height: 70px;
    display: flex;
    align-items: center;
    justify-content: space-between;
}
```

```
/* Brand */
.nav-brand {
    display: flex;
    align-items: center;
    font-size: 1.6rem;
    font-weight: 700;
    color: #667eea;
    gap: 8px;
    text-decoration: none;
}
.nav-brand i {
    font-size: 1.9rem;
    color: #667eea;
}
/* Menu */
.nav-menu {
    display: flex;
    gap: 25px;
}
.nav-link {
    text-decoration: none;
    color: #333;
    font-weight: 500;
    position: relative;
    padding-bottom: 3px;
    transition: color 0.3s ease;
}
.nav-link:hover,
.nav-link.active {
    color: #667eea;
}
.nav-link::after {
    content: "";
    position: absolute;
    left: 0;
    bottom: -4px;
    width: 0;
    height: 2px;
    background: #667eea;
    transition: width 0.3s ease;
}
.nav-link:hover::after, .nav-link.active::after {
    width: 100%;
}
/* Buttons */
.btn {
```

```
    padding: 8px 16px;
    border-radius: 6px;
    border: none;
    font-weight: 500;
    cursor: pointer;
    transition: all 0.3s ease;
    font-size: 0.95rem;
}
.btn-primary {
    background: linear-gradient(135deg, #667eea, #764ba2);
    color: #fff;
}
.btn-primary:hover {
    transform: translateY(-2px);
    box-shadow: 0 6px 15px rgba(102, 126, 234, 0.3);
}
.btn-outline {
    background: transparent;
    border: 2px solid #667eea;
    color: #667eea;
}
.btn-outline:hover {
    background: #667eea;
    color: #fff;
}
/* ===== MAIN CONTENT ===== */
main {
    padding-top: 100px; /* space for fixed navbar */
}
/* HERO SECTION */
.hero-title {
    font-size: 2.8rem;
    font-weight: 700;
    color: #333;
    margin-bottom: 20px;
}
.hero-subtitle {
    font-size: 1.1rem;
    color: #555;
    margin-bottom: 30px;
}
/* FEATURES / CARDS */
.features-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(260px, 1fr));
    gap: 30px;
```

```
    max-width: 1000px;
    margin: auto;
}
.feature-card {
    background: #fff;
    border-radius: 12px;
    padding: 35px 25px;
    text-align: center;
    box-shadow: 0 6px 18px rgba(0, 0, 0, 0.08);
    transition: transform 0.3s ease, box-shadow 0.3s ease;
    cursor: pointer;
}
.feature-card:hover {
    transform: translateY(-5px);
    box-shadow: 0 12px 25px rgba(0, 0, 0, 0.1);
}
.feature-icon {
    font-size: 2.8rem;
    color: #667eea;
    margin-bottom: 20px;
}
.feature-card h3 {
    font-size: 1.35rem;
    color: #333;
    margin-bottom: 12px;
}
.feature-card p {
    font-size: 0.95rem;
    color: #555;
    line-height: 1.5;
}
/* DASHBOARD WELCOME */
.dashboard-welcome {
    text-align: center;
    margin-bottom: 40px;
}
.dashboard-welcome h1 {
    font-size: 2.5rem;
    color: #333;
}
.dashboard-welcome p {
    font-size: 1.1rem;
    color: #555;
    margin-top: 12px;
}
/* RECENT ACTIVITY */
```

```
.recent-activity {
    max-width: 900px;
    margin: 50px auto 0;
    background: #fff;
    padding: 25px 20px;
    border-radius: 12px;
    box-shadow: 0 6px 18px rgba(0, 0, 0, 0.08);
}
.recent-activity h2 {
    font-size: 1.7rem;
    color: #333;
    margin-bottom: 15px;
}
.recent-activity p {
    font-size: 0.95rem;
    color: #555;
}
/* FOOTER */
footer {
    background: #667eea;
    color: #fff;
    padding: 40px 20px;
    text-align: center;
    font-size: 0.95rem;
}
```

Admin_dashboard.php

```
<?php
session_start();
if(!isset($_SESSION['user_logged_in']) || $_SESSION['user_role'] !== 'admin'){
    header("Location: ../login.php");
    exit();
}

include '../db.php';

// Fetch summary counts safely
$total_users_result = mysqli_query($conn, "SELECT COUNT(*) AS count FROM users");
$total_users_row = mysqli_fetch_assoc($total_users_result);
$total_users = $total_users_row['count'];

$total_books_result = mysqli_query($conn, "SELECT COUNT(*) AS count FROM books");
$total_books_row = mysqli_fetch_assoc($total_books_result);
$total_books = $total_books_row['count'];
```

```
$total_purchases_result = mysqli_query($conn, "SELECT COUNT(*) AS count FROM
purchases");
$total_purchases_row = mysqli_fetch_assoc($total_purchases_result);
$total_purchases = $total_purchases_row['count'];

$total_notes_result = mysqli_query($conn, "SELECT COUNT(*) AS count FROM notes");
$total_notes_row = mysqli_fetch_assoc($total_notes_result);
$total_notes = $total_notes_row['count'];

// Fetch users for table
$users = mysqli_query($conn, "SELECT * FROM users ORDER BY created_at DESC");
?>
<html>
<body>
<nav class="navbar">
   <div class="nav-container">
      <div class="nav-brand"><i class="fas fa-book-open"></i> LittleHUB</div>
      <div class="nav-brand"><i class="fas fa-user-shield"></i> Admin Panel</div>
      <div class="nav-menu">
         <a href="admin_dashboard.php" class="nav-link active">Dashboard</a>
         <a href="admin_users.php" class="nav-link">Users</a>
         <a href="admin_books.php" class="nav-link">Books</a>
         <a href="admin_purchases.php" class="nav-link">Purchases</a>
         <a href="admin_notes.php" class="nav-link">Notes</a>
         <a href="../logout.php" class="nav-link">Logout</a>
      </div>
   </div>
</nav>
<main>
<h1>Admin Dashboard</h1>
<div class="stats">
   <div class="stat-box"><h2><?php echo $total_users; ?></h2><p>Total Users</p></div>
   <div class="stat-box"><h2><?php echo $total_books; ?></h2><p>Total Books</p></div>
   <div class="stat-box"><h2><?php echo $total_purchases; ?></h2><p>Total
Purchases</p></div>
   <div class="stat-box"><h2><?php echo $total_notes; ?></h2><p>Total Notes</p></div>
</div>
<h2>All Users</h2>
<a href="add_user.php" class="btn edit">+ Add New User</a>
<table>
<thead>
<tr>
<th>ID</th>
<th>Name</th>
<th>Email</th>
<th>Role</th>
```

```
<th>Created At</th>
<th>Actions</th>
</tr>
</thead>
<tbody>
<?php while($user = mysqli_fetch_assoc($users)): ?>
<tr>
<td><?php echo $user['user_id']; ?></td>
<td><?php echo htmlspecialchars($user['name']); ?></td>
<td><?php echo htmlspecialchars($user['email']); ?></td>
<td><?php echo $user['role']; ?></td>
<td><?php echo $user['created_at']; ?></td>
<td>
<a href="edit_user.php?id=<?php echo $user['user_id']; ?>" class="btn edit">Edit</a>
<a href="delete_user.php?id=<?php echo $user['user_id']; ?>" class="btn delete" onclick="return
confirm('Are you sure you want to delete this user?')">Delete</a>
</td>
</tr>
<?php endwhile; ?>
</tbody>
</table>
</main>
</body>
</html>
<?php mysqli_close($conn); ?>
```

Admin_notes.php

```
<?php
session_start();
if(!isset($_SESSION['user_logged_in']) || $_SESSION['user_role'] !== 'admin'){
    header("Location: ../login.php");
    exit();
}
include '../db.php';
$message = "";
// Handle file upload
if($_SERVER['REQUEST_METHOD'] === 'POST'){
    $subject = mysqli_real_escape_string($conn, $_POST['subject']);
    $title = mysqli_real_escape_string($conn, $_POST['title']);
    $description = mysqli_real_escape_string($conn, $_POST['description']);
    if(isset($_FILES['file']) && $_FILES['file']['error'] === 0){
        $filename = time() . '_' . basename($_FILES['file']['name']);
        $filepath = '../notes_pdf/' . $filename;
        if(move_uploaded_file($_FILES['file']['tmp_name'], $filepath)){
            $uploaded_by = $_SESSION['user_id'];
            $sql = "INSERT INTO notes (subject, title, description, file_path, uploaded_by)
```

```
                VALUES ('$subject', '$title', '$description', '$filename', $uploaded_by)";
            if(mysqli_query($conn, $sql)){
                $message = "Note uploaded successfully.";
            } else {
                $message = "Database error: " . mysqli_error($conn);
            }
        } else {
            $message = "Failed to upload file.";
        }
    } else {
        $message = "Please select a file to upload.";
    }
}
// Fetch all notes
$notes = mysqli_query($conn, "SELECT n.note_id, n.subject, n.title, n.file_path, n.uploaded_at,
u.name
                        FROM notes n
                        JOIN users u ON n.uploaded_by=u.user_id
                        ORDER BY n.uploaded_at DESC");
?>
<html>
<body>
<nav class="navbar">
    <div class="nav-container">
        <div class="nav-brand"><i class="fas fa-book-open"></i> LittleHUB</div>
        <div class="nav-brand"><i class="fas fa-user-shield"></i> Admin Panel</div>
        <div class="nav-menu">
            <a href="admin_dashboard.php" class="nav-link">Dashboard</a>
            <a href="admin_users.php" class="nav-link">Users</a>
            <a href="admin_books.php" class="nav-link">Books</a>
            <a href="admin_purchases.php" class="nav-link">Purchases</a>
            <a href="admin_notes.php" class="nav-link active">Notes</a>
            <a href="../logout.php" class="nav-link">Logout</a>
        </div>
    </div>
</nav>
<main>
<h1>Notes Management</h1>
<?php if($message!="") echo "<p style='color:red;'>$message</p>"; ?>
<h2>Upload New Note</h2>
<form action="" method="POST" enctype="multipart/form-data">
    <label>Subject:</label>
    <input type="text" name="subject" required>
    <label>Title:</label>
    <input type="text" name="title" required>
    <label>Description:</label>
```

```
    <textarea name="description" rows="3"></textarea>
    <label>File (PDF):</label>
    <input type="file" name="file" accept="application/pdf" required>
    <button type="submit">Upload Note</button>
</form>
<h2>All Notes</h2>
<table>
<thead>
<tr>
<th>ID</th>
<th>Subject</th>
<th>Title</th>
<th>Uploaded By</th>
<th>Uploaded At</th>
<th>File</th>
<th>Actions</th>
</tr>
</thead>
<tbody>
<?php while($note = mysqli_fetch_assoc($notes)): ?>
<tr>
<td><?php echo $note['note_id']; ?></td>
<td><?php echo htmlspecialchars($note['subject']); ?></td>
<td><?php echo htmlspecialchars($note['title']); ?></td>
<td><?php echo htmlspecialchars($note['name']); ?></td>
<td><?php echo $note['uploaded_at']; ?></td>
<td><a href="../notes_pdf/<?php echo $note['file_path']; ?>" target="_blank">View</a></td>
<td>
    <a href="edit_note.php?id=<?php echo $note['note_id']; ?>" class="btn edit">Edit</a>
    <a href="delete_note.php?id=<?php echo $note['note_id']; ?>" class="btn delete"
onclick="return confirm('Are you sure you want to delete this note?');">Delete</a>
</td>
</tr>
<?php endwhile; ?>
</tbody>
</table>
</main>
</body>
</html>
<?php mysqli_close($conn); ?>
```

Admin_books.php

```
<?php
session_start();
if(!isset($_SESSION['user_logged_in']) || $_SESSION['user_role'] !== 'admin'){
    header("Location: ../login.php");
```

```
    exit();
}
include '../db.php';
$message = "";
// Handle delete action
if(isset($_GET['delete_id'])){
    $delete_id = intval($_GET['delete_id']);

    // Get the book's cover and pdf to delete from server
    $book_res = mysqli_query($conn, "SELECT * FROM books WHERE book_id=$delete_id");
    $book = mysqli_fetch_assoc($book_res);
    if($book){
        if(!empty($book['cover']) && file_exists("../cover/" . $book['cover'])){
            unlink("../cover/" . $book['cover']);
        }
        if(!empty($book['pdf']) && file_exists("../books_pdf/" . $book['pdf'])){
            unlink("../books_pdf/" . $book['pdf']);
        }
    }
    mysqli_query($conn, "DELETE FROM books WHERE book_id=$delete_id");
    header("Location: admin_books.php");
    exit();
}
// Fetch all books
$books = mysqli_query($conn, "SELECT * FROM books ORDER BY created_at DESC");
?>
<html>
<body>
<!-- NAVBAR -->
<nav class="navbar">
    <div class="nav-container">
        <div class="nav-brand"><i class="fas fa-book-open"></i> LittleHUB</div>
        <div class="nav-brand"><i class="fas fa-user-shield"></i> Admin Panel</div>
        <div class="nav-menu">
            <a href="admin_dashboard.php" class="nav-link">Dashboard</a>
            <a href="admin_users.php" class="nav-link">Users</a>
            <a href="admin_books.php" class="nav-link active">Books</a>
            <a href="admin_purchases.php" class="nav-link">Purchases</a>
            <a href="admin_notes.php" class="nav-link">Notes</a>
            <a href="../logout.php" class="nav-link">Logout</a>
        </div>
    </div>
</nav>
<main style="padding:120px 20px 50px 20px;">
<h1>Books Management</h1>
<a href="add_book.php" class="add-btn">+ Add New Book</a>
```

```
<table>
<thead>
<tr>
<th>ID</th>
<th>Title</th>
<th>Author</th>
<th>Price</th>
<th>Cover</th>
<th>PDF</th>
<th>Created At</th>
<th>Actions</th>
</tr>
</thead>
<tbody>
<?php while($book = mysqli_fetch_assoc($books)): ?>
<tr>
<td><?php echo $book['book_id']; ?></td>
<td><?php echo htmlspecialchars($book['title']); ?></td>
<td><?php echo htmlspecialchars($book['author']); ?></td>
<td><?php echo number_format($book['price'],2); ?></td>
<td>
<?php
if(!empty($book['cover']) && file_exists("../cover/".$book['cover'])){
   echo "<img src='../cover/{$book['cover']}' class='cover-thumb' alt='cover'>";
} else {
   echo "N/A";
}
?>
</td>
<td>
<?php
if(!empty($book['pdf']) && file_exists("../books_pdf/".$book['pdf'])){
   echo "<a href='../books_pdf/{$book['pdf']}' target='_blank'>View PDF</a>";
} else {
   echo "N/A";
}
?>
</td>
<td><?php echo $book['created_at']; ?></td>
<td>
<a href="edit_book.php?id=<?php echo $book['book_id']; ?>" class="btn edit">Edit</a>
<a href="admin_books.php?delete_id=<?php echo $book['book_id']; ?>" class="btn delete"
onclick="return confirm('Are you sure you want to delete this book?');">Delete</a>
</td>
</tr>
<?php endwhile; ?>
```

```
</tbody>
</table>
</main>
</body>
</html>
<?php mysqli_close($conn); ?>
```

Dashboard.php

```
<?php
session_start();
if (!isset($_SESSION['user_logged_in']) || $_SESSION['user_logged_in'] !== true) {
    header("Location: login.php");
    exit();
}
include 'db.php';
$user_id = $_SESSION['user_id'];
$user_name = $_SESSION['user_name'];
// Fetch purchased books
$purchased_books_query = "
SELECT b.* FROM books b
JOIN purchases p ON b.book_id = p.book_id
WHERE p.user_id = $user_id
ORDER BY p.purchased_at DESC
";
$purchased_books = mysqli_query($conn, $purchased_books_query);
?>
<!DOCTYPE html>
<html lang="en">
<body>
<nav class="navbar">
    <div class="nav-container">
        <div class="nav-brand">
            <i class="fas fa-book-open"></i> LittleHUB
        </div>
        <div class="nav-menu">
            <a href="dashboard.php" class="nav-link active">Dashboard</a>
            <a href="books.php" class="nav-link">Books</a>
            <a href="notes.php" class="nav-link">Notes</a>
            <a href="store.php" class="nav-link">Store</a>
            <a href="aboutus.php" class="nav-link">About Us</a>
        </div>
        <div class="nav-actions">
            <button class="btn btn-primary"
onclick="window.location.href='logout.php'">Logout</button>
        </div>
    </div>
```

```
</nav>
<main style="padding:100px 20px;">
<h1>Welcome, <u><?php echo htmlspecialchars($user_name); ?></u>!</h1>
<p>Here’s your dashboard to access all LittleHUB features.</p>
<div class="features-grid">
   <div class="feature-card" onclick="window.location.href='books.php'">
      <div class="feature-icon"><i class="fas fa-book-reader"></i></div>
      <h3>Online Book Reading</h3>
      <p>Read your favorite books online with our advanced reader interface.</p>
   </div>
   <div class="feature-card" onclick="window.location.href='store.php'">
      <div class="feature-icon"><i class="fas fa-shopping-cart"></i></div>
      <h3>Buy Books</h3>
      <p>Purchase digital or physical books from our store easily.</p>
   </div>
   <div class="feature-card" onclick="window.location.href='notes.php'">
      <div class="feature-icon"><i class="fas fa-download"></i></div>
      <h3>Download College Notes</h3>
      <p>Download notes for your subjects uploaded by other students and teachers.</p>
   </div>
</div>
<!-- Purchased Books Section -->
<section style="margin-top:50px;">
   <h2>Your Purchased Books</h2>
   <?php if(mysqli_num_rows($purchased_books) > 0): ?>
      <div class="features-grid">
         <?php while($book = mysqli_fetch_assoc($purchased_books)): ?>
            <div class="purchased-card">
               <div class="purchased-icon"><i class="fas fa-book"></i></div>
               <h3><?php echo htmlspecialchars($book['title']); ?></h3>
               <p>Author: <?php echo htmlspecialchars($book['author']); ?></p>
               <p>Category: <?php echo htmlspecialchars($book['category']); ?></p>
               <div class="action-buttons">
                  <?php if(!empty($book['file_path'])): ?>
                     <a href="download.php?book_id=<?php echo $book['book_id']; ?>"
class="btn-download">
                        <i class="fas fa-download"></i> Download
                     </a>
                     <a href="books.php?book_id=<?php echo $book['book_id']; ?>"
class="btn-view" target="_blank">
                        <i class="fas fa-eye"></i> View Online
                     </a>
                  <?php endif; ?>
               </div>
            </div>
         <?php endwhile; ?>
```

```
        </div>
    <?php else: ?>
        <p style="font-size:1rem; color:#555;">You haven't purchased any books yet.</p>
    <?php endif; ?>
</section>
</main>
</body>
</html>
```

Store.php

```
<?php
session_start();
if (!isset($_SESSION['user_logged_in']) || $_SESSION['user_logged_in'] !== true) {
    header("Location: login.php"); // redirect to login if not logged in
    exit();
}
$user_name = $_SESSION['user_name'];
// Database connection
$conn = mysqli_connect("localhost", "root", "", "lithub_db");
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}
// Fetch books from the database
$sql = "SELECT * FROM books";
$result = mysqli_query($conn, $sql);

if(!$result){
    die("Query Failed: " . mysqli_error($conn));
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>LittleHUB - Store</title>
<link rel="stylesheet" href="css/style.css">
<link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/5.15.4/css/all.min.css">
</head>
<body>
<nav class="navbar">
    <div class="nav-container">
        <div class="nav-brand">
            <i class="fas fa-book-open"></i>
            <span>LittleHUB</span>
```

```
        </div>
        <div class="nav-menu">
            <a href="dashboard.php" class="nav-link">Dashboard</a>
            <a href="books.php" class="nav-link">Books</a>
            <a href="notes.php" class="nav-link">Notes</a>
            <a href="store.php" class="nav-link active">Store</a>
            <a href="aboutus.php" class="nav-link">About Us</a>
        </div>
        <div class="nav-actions">
            <button class="btn btn-primary"
onclick="window.location.href='logout.php'">Logout</button>
        </div>
    </div>
</nav>
<main style="padding:100px 20px;">
    <h1 class="hero-title">Explore Our Store</h1>
    <p style="font-size:1.1rem; margin-bottom:30px;">Explore digital and physical books
available for purchase.</p>

    <div class="features-grid" style="max-width:1200px; margin:auto;">
        <?php
        if(mysqli_num_rows($result) > 0){
            while($book = mysqli_fetch_assoc($result)){
                ?>
                <div class="feature-card">
                    <div class="feature-icon">
                        <i class="fas fa-book"></i>
                    </div>
                    <h3><?php echo htmlspecialchars($book['title']); ?></h3>
                    <p>Author: <?php echo htmlspecialchars($book['author']); ?></p>
                    <p>Category: <?php echo htmlspecialchars($book['category']); ?></p>
                    <p>Price: ₹<?php echo htmlspecialchars($book['price']); ?></p>
                    <form method="POST" action="buy.php">
                        <input type="hidden" name="book_id" value="<?php echo $book['book_id'];
?>">
                        <button type="submit" class="btn btn-primary" style="margin-top:10px;">Buy
Now</button>
                    </form>
                </div>
                <?php
            }
        } else {
            echo "<p>No books available at the moment.</p>";
        }
        ?>
    </div>
```

```
</main>
</body>
</html>
<?php
mysqli_close($conn);
?>
```

Books.php

```
<?php
session_start();
include 'db.php';
if (!isset($_SESSION['user_id'])) {
   header("Location: login.php");
   exit();
}
$user_name = $_SESSION['user_name'];
?>
<!DOCTYPE html>
<html lang="en"><body>
<nav class="navbar" id="navbar">
   <div class="nav-container">
      <div class="nav-brand">
         <i class="fas fa-book-open"></i>
         <span>LittleHUB</span>
      </div>
      <div class="nav-menu" id="navMenu">
         <a href="dashboard.php" class="nav-link">Dashboard</a>
         <a href="books.php" class="nav-link active">Books</a>
         <a href="notes.php" class="nav-link">Notes</a>
         <a href="store.php" class="nav-link">Store</a>
         <a href="aboutus.php" class="nav-link">About Us</a>
      </div>
      <div class="nav-actions">
         <button class="btn btn-primary"
onclick="window.location.href='logout.php'">Logout</button>
      </div>
   </div>
</nav>
<main style="padding: 100px 20px;">
   <h1 class="hero-title">Explore Our Store!!!!</h1>
   <p style="margin-bottom:30px;">Explore digital and physical books available for
purchase.</p>
   <div class="features-grid" style="max-width:1000px; margin:auto; display:grid;
grid-template-columns: repeat(auto-fill,minmax(220px,1fr)); gap:20px;">
      <?php
      $query = "SELECT * FROM books ORDER BY created_at DESC";
```

```
        $result = mysqli_query($conn, $query);

        if($result && mysqli_num_rows($result) > 0){
            while($book = mysqli_fetch_assoc($result)){
                echo '<div class="book-card">';
                // Cover Image
                $cover = !empty($book['cover_image']) ? htmlspecialchars($book['cover_image']) :
'covers/default_cover.jpg';
                echo '<img class="book-cover" src="'.$cover.'"
alt="'.htmlspecialchars($book['title']).'">';
                // Title & Author
                echo '<h3>'.htmlspecialchars($book['title']).'</h3>';
                echo '<p><strong>Author:</strong> '.htmlspecialchars($book['author']).'</p>';
                echo '<p><strong>Category:</strong> '.htmlspecialchars($book['category']).'</p>';
                // Price
                echo '<p class="price">₹'.number_format($book['price'],2).'</p>';
                // Sample PDF button
                if(!empty($book['file_path'])){
                    $pdf = htmlspecialchars($book['file_path']);
                    echo '<button class="btn btn-sample"
onclick="window.open(\'.$pdf.'\',\'_blank\')">Read Sample</button>';
                }
                // Buy button
                echo '<button class="btn btn-primary btn-buy">Buy Now</button>';
                echo '</div>';
            }
        } else {
            echo '<p>No books available at the moment.</p>';
        }
        ?>
    </div>
</main>
</body>
</html>
```

Notes.php

```
<?php
session_start();
if (!isset($_SESSION['user_logged_in']) || $_SESSION['user_logged_in'] !== true) {
    header("Location: login.php");
    exit();
}
$user_name = $_SESSION['user_name'];
include 'db.php';
// Fetch notes
$notes = mysqli_query($conn, "SELECT * FROM notes ORDER BY uploaded_at DESC");
```

```
// Dynamically get the base URL
$protocol = (!empty($_SERVER['HTTPS']) && $_SERVER['HTTPS'] !== 'off' ||
$_SERVER['SERVER_PORT'] == 443) ? "https://" : "http://";
$baseURL = $protocol . $_SERVER['HTTP_HOST'] . dirname($_SERVER['PHP_SELF']) . '/';
?>
<!DOCTYPE html>
<html lang="en">
<body>
<!-- NAVBAR -->
<nav class="navbar">
   <div class="nav-container">
      <div class="nav-brand">
         <i class="fas fa-book-open"></i>
         <span>LittleHUB</span>
      </div>
      <div class="nav-menu">
         <a href="dashboard.php" class="nav-link">Dashboard</a>
         <a href="books.php" class="nav-link">Books</a>
         <a href="notes.php" class="nav-link active">Notes</a>
         <a href="store.php" class="nav-link">Store</a>
         <a href="aboutus.php" class="nav-link">About Us</a>
      </div>
      <div class="nav-actions">
         <button class="btn btn-primary"
onclick="window.location.href='logout.php'">Logout</button>
      </div>
   </div>
</nav>
<main id="mainContent" style="padding: 80px 20px;">
   <div class="dashboard-welcome" style="text-align:center; margin-bottom:40px;">
      <h1>Welcome, <u><?php echo htmlspecialchars($user_name); ?></u>!</h1>
      <p>Download and share college notes uploaded by students and teachers.</p>
   </div>
   <div class="features-grid">
      <?php
      if(mysqli_num_rows($notes) > 0) {
         while($note = mysqli_fetch_assoc($notes)) {
            $title = htmlspecialchars($note['title']);
            $subject = htmlspecialchars($note['subject']);
            $filePath = htmlspecialchars($note['file_path']);

            // Use dynamic base URL
            $fullURL = $baseURL . $filePath;
            $encodedURL = urlencode($fullURL);
            $encodedTitle = urlencode("Check out this note: $title");
```

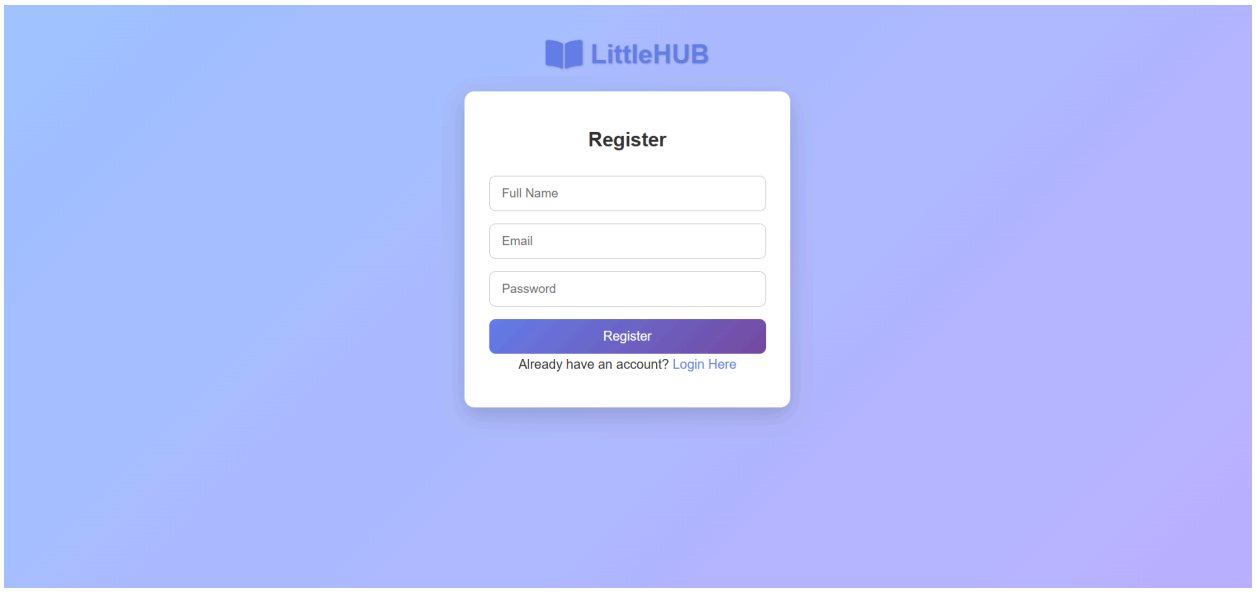
```
            $whatsappLink =
"https://api.whatsapp.com/send?text=$encodedTitle%20$encodedURL";
            $emailLink = "mailto:?subject=".rawurlencode("Sharing Note: $title").
                    "&body=".rawurlencode("Hi,%0D%0A%0D%0ACheck out this note:
$title%0D%0ASubject: $subject%0D%0ADownload here: $fullURL");
            ?>
            <div class="note-card">
               <div class="note-icon"><i class="fas fa-file-pdf"></i></div>
               <h3><?php echo $title; ?></h3>
               <p><strong>Subject:</strong> <?php echo $subject; ?></p>
               <div class="action-buttons">
                  <!-- View PDF -->
                  <a href="<?php echo $fullURL; ?>" class="btn-download" target="_blank">
                     <i class="fas fa-eye"></i> View
                  </a>
                  <!-- Download PDF -->
                  <a href="<?php echo $filePath; ?>" class="btn-download" download>
                     <i class="fas fa-download"></i> Download
                  </a>
                  <!-- Share via WhatsApp -->
                  <a href="<?php echo $whatsappLink; ?>" class="btn-whatsapp" target="_blank">
                     <i class="fab fa-whatsapp"></i> Share on WhatsApp
                  </a>
                  <!-- Share via Email -->
                  <a href="<?php echo $emailLink; ?>" class="btn-email">
                     <i class="fas fa-envelope"></i> Share via Email
                  </a>
               </div>
            </div>
      <?php }
      } else { ?>
         <p style="text-align:center; font-size:1.1rem; color:#555;">No notes available yet.</p>
      <?php } ?>
   </div>
</main>
</body>
</html>
```

Logout.php

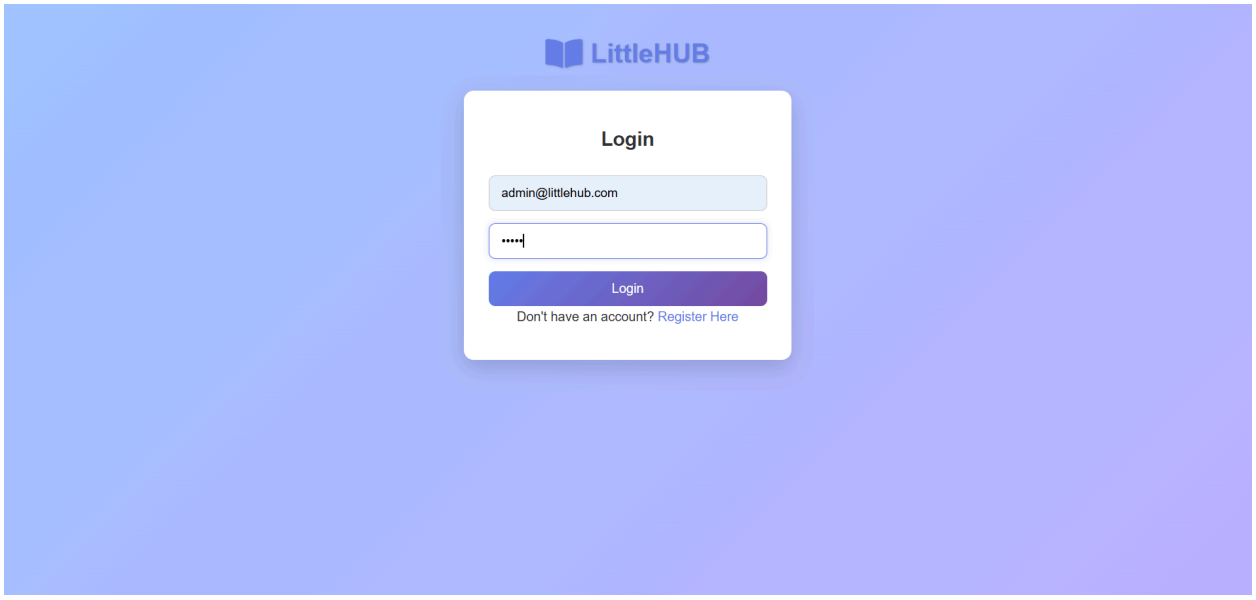
```
<?php
session_start();
session_unset();
session_destroy();
header("Location: index.php");
exit();
?>
```

OUTPUT SCREENS

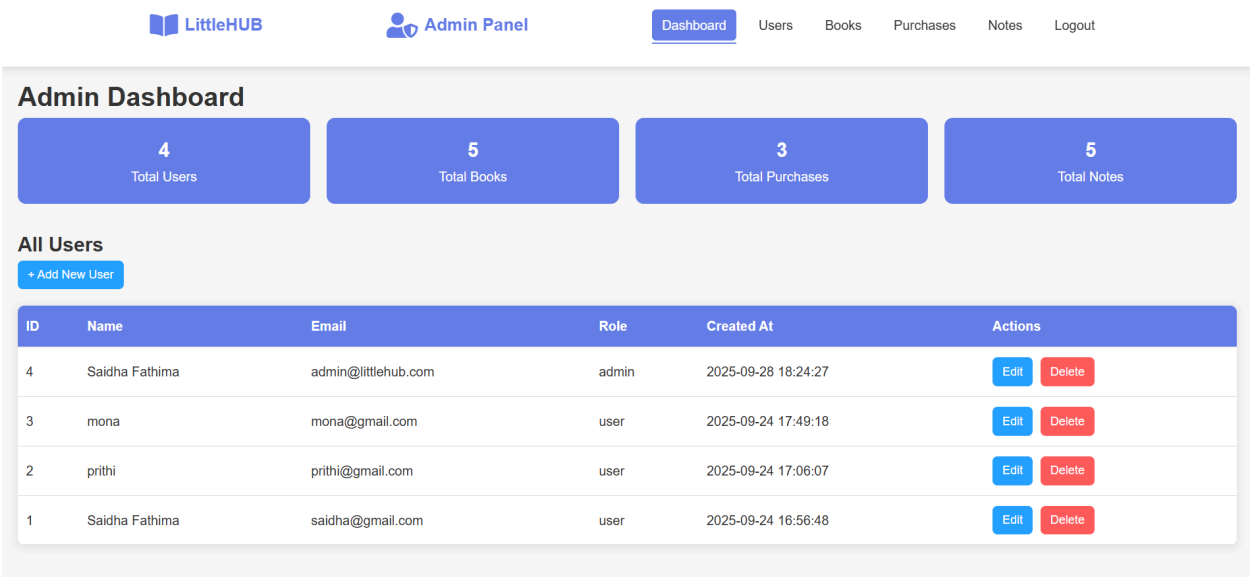
Register page:



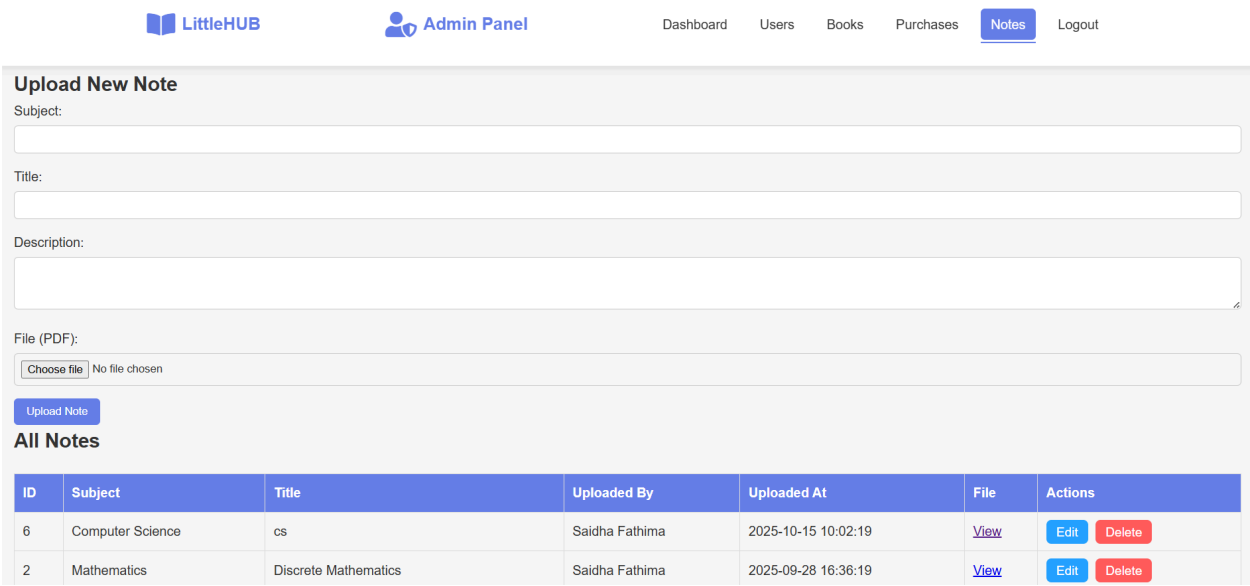
Login Page:



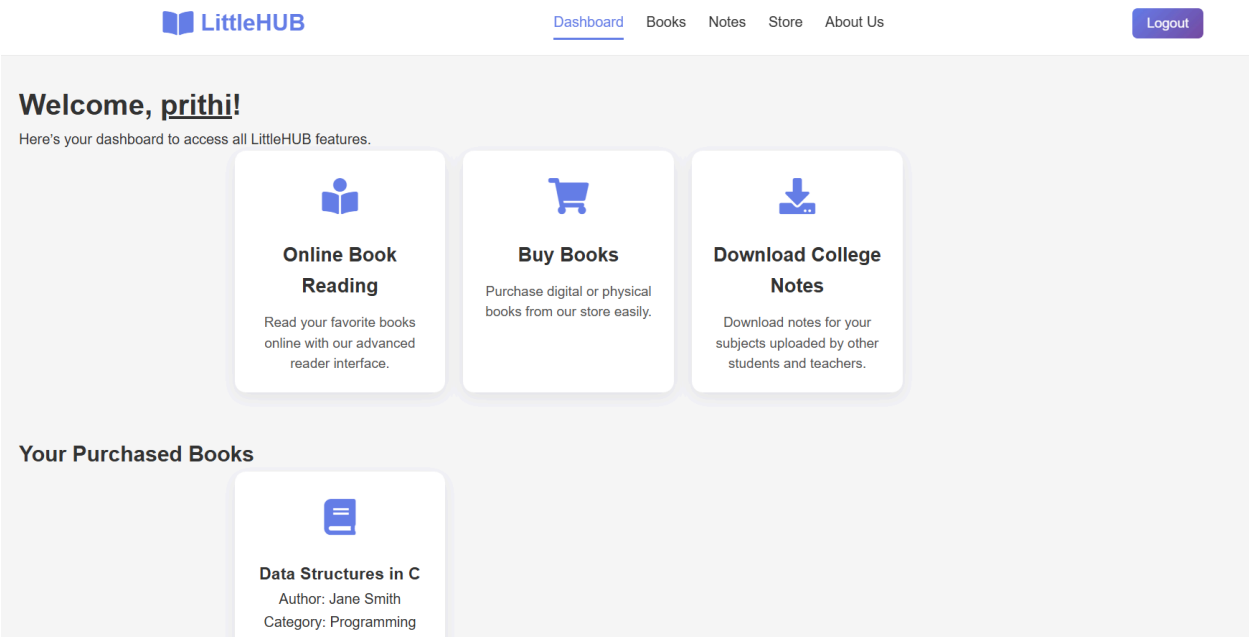
Admin Dashboard page:



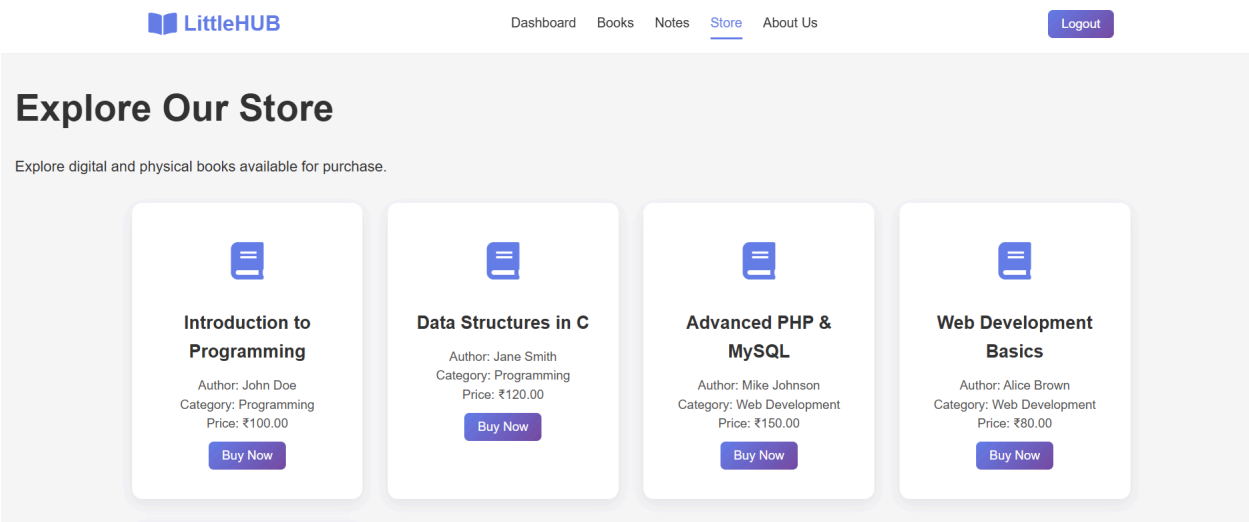
Admin notes page:



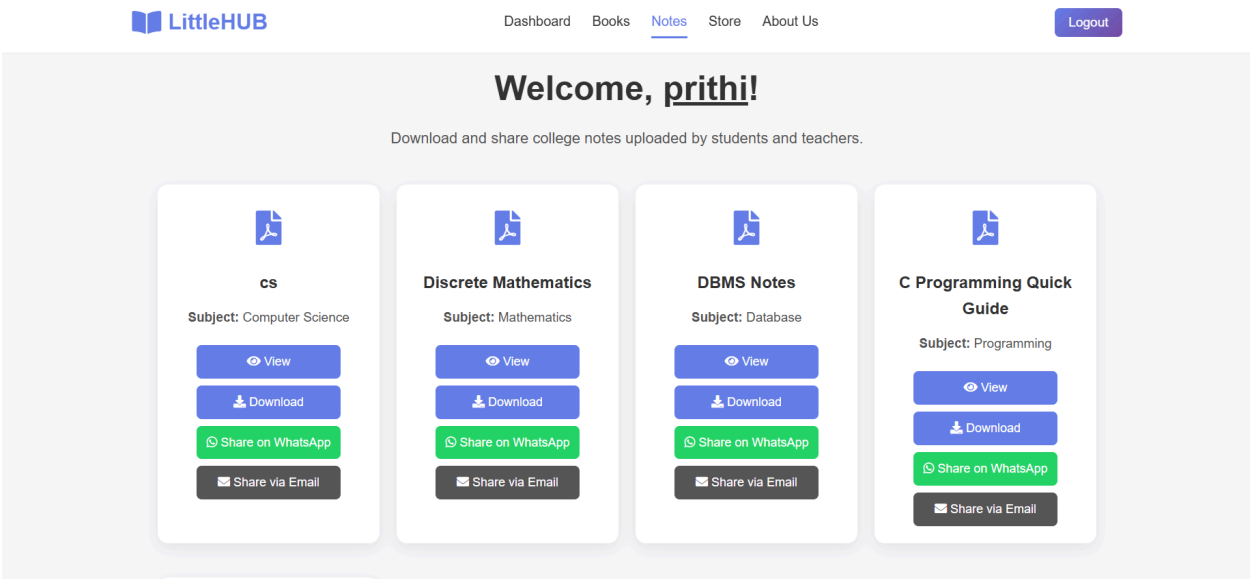
User dashboard page:



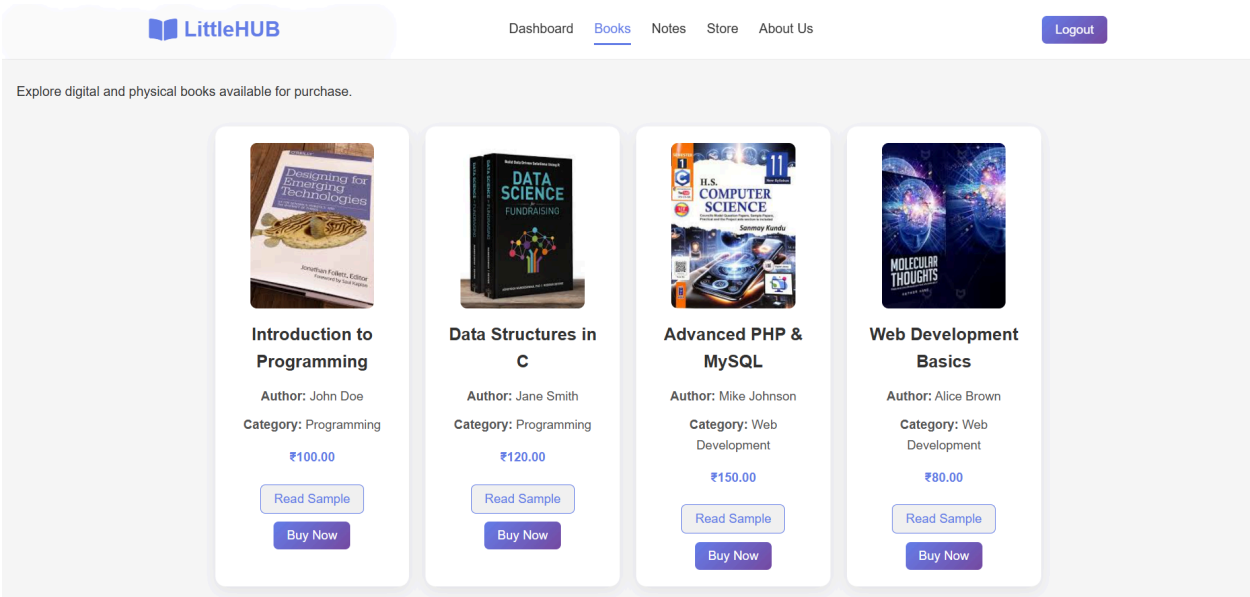
Store page:



Notes page:



Books page:



CONCLUSION

The LittleHub project successfully demonstrates the development of a comprehensive, database-driven web application that integrates essential functionalities such as user authentication, digital resource management, and secure file handling. It provides a centralized platform for students and educators to share notes, manage books, and access learning materials efficiently within a user-friendly interface.

By implementing modules for book management, note uploads, and purchase tracking, the project showcases the practical application of PHP and MySQL in real-world scenarios. Additionally, the use of a responsive and consistent design theme ensures a smooth and engaging user experience across all devices.

Throughout the development of LittleHub, various technical concepts such as CRUD operations, session handling, database relationships, and file uploads were successfully implemented, enhancing both the functionality and reliability of the system.

In conclusion, LittleHub fulfills its primary objective — to serve as a digital learning hub that encourages collaboration, simplifies academic resource sharing, and supports the transition toward modern, technology-driven education systems. With further enhancements such as cloud integration, API support, and AI-based recommendations, LittleHub has strong potential for future expansion into a more powerful and intelligent e-learning ecosystem.

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FEATURE ENHANCEMENTS

The LittleHub project, though currently offering robust functionality for managing books, notes, users, and purchases, can be further enhanced with the following features to improve usability, scalability, and intelligence:

1. AI-based book recommendations to suggest books and notes using machine learning.
2. Cloud storage integration for saving uploads on Google Drive, Dropbox, or AWS S3.
3. Advanced search and filters for easy book and note discovery.
4. User activity analytics dashboard to track user activity and downloads.
5. Mobile application integration for Android and iOS users.
6. E-book reader integration to read books directly in the browser.
7. Subscription and payment gateway for premium features and secure payments.
8. Automated email notifications for updates and offers.
9. Multi-language support for wider accessibility.
10. Enhanced security features with two-factor authentication and reCAPTCHA.